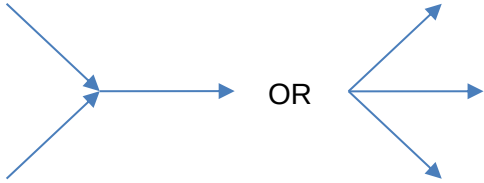


Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
3	(a)		Vector diagram roughly as shown (1)  Vertical components cancel (1) Accept forces due to +ve charges cancel Field is right OR attractive force towards positives (1) Electron is negative linked to F (or a) left OR repulsive from -ve (1)		4		4	2	
	(b)		Use of $F = \frac{kQq}{r^2}$ OR $F = EQ$ and $E = \frac{kq}{r^2}$ (1) (expect to see 9.6×10^{-18} [N]) $2E\cos 60 + E$ or $2E\sin 30 + E$ OR $2F\cos 60 + F$ or $2F\sin 30 + F$ (1) (expect to see 1.92×10^{-17} [N] or 120 [N C⁻¹]) Use of $F = ma$ (1) (can be awarded for any valid force $\div 9.11 \times 10^{-31}$) Correct answer = 2.1×10^{13} [m s⁻²] (no ecf for this, must be correct) (1)	1	1		4	3	